DATA ITEM DESCRIPTION

Form Approved
OMB No. 0704.0188

1. TITLE	2. IDENTIFICATION NUMBER
NUCLEAR SURVIVABILITY TEST REPORT	DI-NUOR-80929A

3. DESCRIPTION/PURPOSE

3.1 The Nuclear Survivability Test Report shall document the results of nuclear survivability test performed on systems, subsystems, structures, circuits, components, materials, and piece parts and provide the engineering evaluation of the system's nuclear survivability.

4. APPROVAL DATE	5. OFFICE OF PRIMARY RESPONSIBILITY (OPR)	6a. DTIC REQUESTED	6b. GIDEP APPLICABLE
920604	A/LABCOM/SLCHD-NW-P		

7. APPLICATION/INTERRELATIONSHIP

7.1 Application. When solicitation or contract contains a requirement to demonstrate compliance with nuclear survivability criteria, this Data Item Description (DID) shall be listed on the Contract Data Requirements List (DD Form 1423).

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8. APPROVAL LIMITATION	9a. APPLICABLE FORMS	9b. AMSC NUMBER
		A6740

10. PREPARATION INSTRUCTIONS

10.1 Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

10.2 General. The Nuclear Survivability Test Report shall contain all the data and conclusions resulting from the test(s) delineated in the current revisions of the Nuclear Survivability Test Plan and shall include the following information:

10.3 Specific data described in the report.

a. Documentation with raw data of the pre- and post-exposure measurements on the piece parts, materials, components, circuits, structures, subassemblies, assemblies, and systems tests. There shall be particular emphasis on identifying and reporting the test results for the hardness critical items (HCI) or hardness critical (HCP) and the mission essential equipments.

b. The report shall document the test conditions for these measurements, including the ambient conditions, the instrumentation for the measurements, the monitor points, the time relation of the measurements with respect to the test level, modes and conditions of the test sample function during the test level and during the measurements, physical orientations of the test samples to the test environment (for system level tests, these

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11. DISTRIBUTION STATEMENT

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7. APPLICATION/INTERRELATIONSHIP (continued)

7.2 This DID contains the format and content preparation instructions for the data product generated by the specific and discrete task requirements as delineated in the contract.

7.3 When a solicitation or contract has this test report(s) as a requirement, then the Nuclear Survivability Test Plan shall also be listed on the DD Form 1423.

7.4 AR 70-60, Army Nuclear Survivability, contains the Army policy applicable to nuclear survivability. HDL-TR-1882-I, Nuclear Weapons Effects on Army Tactical Systems, Vol. I, Overview, provides a comprehensive overview of nuclear weapons effects. HDL-SR-85-3 provides guidelines for the initial nuclear radiation validation testing of Army equipment. MIL-STDs 883 and 750 contain applicable test methods for microcircuits and discrete semiconductor devices. HDL-CR-81-015-1 is a document that provides more specific information. The HDL documents are available from the Defense Technical Information Center.

7.5 This DID supersedes DI-R-1760A.

7.6 The DD Form 1423 shall also list DASIAC as a recipient of the report. DASIAC serves as a DoD wide archive for nuclear survivability purposes. The mailing address is Kaman Tempo/DASIAC, P.O. Box 1479, 816 State St. (for express/private delivery only), Santa Barbara, CA 93102-1479.

7.7 A separate copy of the Special Data (para. 10.6) shall be provided to DASIAC (para. 7.6) in digital format for inclusion into the Electronic Radiation Response Information Center (ERRIC). Contract ERRIC at (805) 963-6484 for formatting instructions.

10. PREPARATION INSTRUCTIONS (continued)

include the Government and Contractor furnished equipments), time after exposure that the measurements are made and the measurements of the simulated environments.

c. The report shall provide a statement of how the simulated environments were measured; where with respect to the sample the measurements were made; and what devices or instruments were used.

d. A detailed description of the instrumentation calibration procedures and dosimetry shall be provided.

e. Detailed test data sheets for each test level with the concurrence signatures of the Government observers and the contractors shall be provided by the contractor in this report.

f. The report shall list the applicable methods of MIL-STDs 750 and 883 used in the tests on semiconductor piece parts.

10.4 System test operational data described in the report.

a. For system level tests, the report shall specify the system operational steps used to set up the system for tests.

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10. PREPARATION INSTRUCTIONS (continued)

b. The report shall detail the equipment post-exposure status and the procedures used to restore the system to full function after the exposure.

10.5 Data analysis described in the report.

a. The report shall provide an analysis for the acceptability of the test data.

b. As appropriate, a detailed description shall be provided of how the response information on piece parts, materials, components, circuits, structures, subassemblies, and assemblies is analyzed to arrive at an evaluation of the system survivability, to include the operational impact of the system's nuclear response, error budgets, and the specification for the allowable downtime.

c. Details shall be provided of the calculations and analysis for the interpolation or extrapolation of the test data to determine how the system responds to the specified criteria. Computer analysis codes used in the analysis shall be named and referenced. All configurations, vehicles, and enclosures for these equipments shall be included in the nuclear survivability evaluations.

d. In the data analysis discussions, the assumptions, opinions, and judgments made to support the conclusions shall be clearly identified.

10.6 Special data for electronic parts. The test report(s) shall provide the data listed below for the neutron, ionizing radiation dose, and ionizing dose-rate responses of electronic pieceparts and shall have the heading Response Data for the Electronics Radiation Response Information Center. A copy of this data is to be provided in digital form as given in paragraph 7.7.

a. Description of device tested.

(1) Type number to include complete MIL-STD part designations, as appropriate (e.g., 2N2930 JTX, M38510/00101BAC), and generic or JEDEC equivalent number.

(2) For non-standard devices, the closest generic or JEDEC part number.

(3) Device class (for example, zener diode, low-power transistor, or linear or digital integrated circuit).

(4) Manufacturer.

(5) Device test sequence identification number.

(6) Manufacturer's lot number, if available, and lot type, i.e., wafer, diffusion or inspection.

(7) Date code or purchase date.

(8) Number of samples tested.

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10. PREPARATION INSTRUCTIONS (continued)

(9) Governing electrical specifications.

(10) Package type.

b. Neutron and ionizing radiation dose effects data.

(1) Pre- and post-irradiation parameters measured, operating points (for example, h_{FE} with I_C and V_{CE} operating points), time between irradiation and measurement, condition of devices between irradiation and measurement (i.e., temperature, mobile bias fixture, time unbiased and applied voltages) and time between consecutive measurements.

(2) Test facility (e.g., reactor, Aberdeen Proving Ground or Febatron 706, HDL....).

(3) Test date.

(4) Bias conditions applied during irradiation.

(5) Irradiation temperature and test temperature.

(6) Test data for each piece part at each radiation level.

(7) Ionizing radiation dose (rad(si)) or neutron fluence (n/cm^2 (1 MeV equivalent damage in Si)). For ionizing radiation dose measurements, the radiation type, dose rate and energy and the duration of the irradiation(s) must be specified.

(8) Applicable MIL-STD methods.

(9) Exposure circuitry and configuration, giving details of materials, distances, designs, and applied voltages.

(10) Parameter and failure measurement apparatus and circuitry.

(11) Dosimetry methods and applicable ASTM standards.

(12) Remarks/comments - relevant information to explain or clarify test conditions or data.

(13) Anomalous incidents or data.

(14) Post-irradiation effects response data for the ionizing radiation dose.

c. Ionizing dose-rate effects data.

(1) Same information as 10.6b, excluding items 7 and 14.

(2) Pulse width, full width half maximum, and the pulse shape.

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10. PREPARATION INSTRUCTIONS (continued)

(3) Photocharge in coulombs or peak primary photocurrent in milliamperes.

(4) Cumulative dose in rad(Si) and dose rate in rad(Si)/second.

(5) As applicable, upset or latchup threshold data.

d. Data Summary

(1) Calculated means and standard deviation for each lot sample.

(2) The aggregate means and standard deviations for the combined lots.

(3) The 99 percent probability, 90 percent confidence limit of response data.

e. General Information

(1) Name and address of company taking data

(2) Person(s) responsible for tests and data records

(3) Phone number for point of contact to include area code and extension.

(4) Any identifiers that will assist in locating data such as lab book numbers.

10.7 Format. The report shall be in the contractor's format.

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